PeriDefT: a periodic-Fourier Transformer for 2D defect formation energy, validated by prospective first-principles DFT

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Abstract

Most published machine-learning surrogates for 2D defect formation energy E_f are benchmarked exclusively on a held-out test fold of the same database used for training, an evaluation that overstates deployment utility because the chemistry of practical interest is out of distribution by construction. We address this gap with three coupled contributions. *First*, we introduce PeriDefT, a 0.75 M-parameter hybrid GNN–Transformer that augments self-attention with a translation-invariant *Periodic Fourier Bias* (PFA) on minimum-image fractional displacements; PFA's exact lattice-translation invariance enables stable joint training across four DFT databases (IMP2D + JARVIS-2D + JARVIS-3D + JARVIS DFT-3D, 26 837 samples) through per-source readout heads. PeriDefT achieves a 4-seed IMP2D test MAE of 0.486 ± 0.025 eV — competitive with ALIGNN (4.03 M, 0.540 eV) at 1/5 the parameter budget — and a 6-seed temperature-calibrated ensemble of 0.443 eV with 90 % interval coverage 93.4 %. *Second*, we run **60 first-principles PBE-DFT calculations** on the model's own out-of-distribution predictions, split into a low-E_f "discovery" bucket and a high-σ "stress-test" bucket. On 37 SCF-converged candidates, the overall predicted–DFT Pearson correlation is +0.354 (95 % bootstrap CI $[+0.06, +0.70]$, excludes zero), and **14/20 = 70 % of bucket-A picks are DFT-confirmed at $E_f < +1$ eV** (95 % CI [50%, 90%]); within bucket B, Pearson(σ_cal, |error|) = −0.288 (95 % CI $[-0.60, +0.09]$), suggesting σ_cal is informative as an in-vs-OOD bucket gate but not as a sample-level reliability score on OOD. *Third*, we release the full pipeline (4-seed and 6-seed ensembles, 287 candidate generation, 60-candidate selection record, 125 Quantum-ESPRESSO SCF outputs) to enable direct reuse for downstream 2D defect screening.

Keywords: 2D materials, defect formation energy, periodic Fourier bias, multi-source training, calibrated uncertainty, prospective DFT validation.

1 Introduction

The defect formation energy

$$E_\mathrm{f}(D,q) = E_\mathrm{tot}(D,q) - E_\mathrm{tot}(\mathrm{host}) - \sum_i n_i\,\mu_i + q\,(E_F + E_\mathrm{VBM}) + E_\mathrm{corr} \tag{1}$$

is the central thermodynamic quantity for defect engineering in two-dimensional materials, but a single first-principles calculation costs tens to hundreds of CPU-hours, prohibitive for high-throughput screening. Existing graph neural networks (CGCNN [8], SchNet [6], ALIGNN [4], M3GNet [3]) and Transformer extensions (Matformer [9], Crystalformer [7], PotNet [5]) reduce this cost by orders of magnitude, but for point defects in 2D the published benchmarks all report held-out-test-fold MAE on the same database used for training. This evaluation says little about the model's deployment utility on out-of-distribution chemistry, which is by construction what every discovery loop encounters.

Contributions.

1. We introduce **PeriDefT**, a 0.75 M-parameter hybrid GNN–Transformer whose self-attention carries a Periodic Fourier Bias (PFA) that is exactly invariant under lattice translations. The exact invariance is what makes the encoder safe to share across geometrically heterogeneous DFT databases via per-source readout heads (IMP2D + JARVIS-2D + JARVIS-3D + JARVIS DFT-3D, 26 837 samples, 3.15× IMP2D). The 4-seed multi-source PeriDefT achieves test MAE 0.486 ± 0.025 eV, an 11 % improvement over a matched single-source baseline; a 6-seed temperature-scaled ensemble reaches 0.443 eV with 90 % interval coverage 93.4 %.
2. We report the first **prospective DFT validation** of an IMP2D-trained 2D-defect E_f model with calibrated uncertainty: 60 PBE-DFT calculations on the model's own OOD predictions, split into a top-μ "discovery" bucket (n=30) and a top-σ_{cal} "stress-test" bucket (n=30). On the 37 SCF-converged candidates, **14/20 = 70%** of bucket-A picks are DFT-confirmed at $E_f < +1$ eV (95 % CI $[50\%, 90\%]$); within bucket B, Pearson$(\sigma_{\text{cal}}, |\text{err}|) = -0.288$ (95 % CI $[-0.60, +0.09]$).
3. We release **open infrastructure**: the trained PeriDefT ensembles, leak-free augmentation contracts (with a self-caught-and-fixed augmentation leak, see App. A.2), the 287-candidate generation script, the 60-candidate selection record, and 125 Quantum-ESPRESSO SCF outputs.

A single-source ablation (Sec. 3) shows that PFA, multi-scale distance bias, and a dual-stream pristine–defect architecture are all statistically tied with the baseline at $N_{\text{train}} = 8\,512$, consistent with the empirical scaling law $\log \text{MAE} \approx 3.39 - 0.40 \log N_{\text{train}} - 0.01 \log N_{\text{params}}$ ($R^2 = 0.95$) of [1]: at fixed corpus size, architectural variation moves log MAE by less than the seed noise. The 11 % gain delivered by multi-source PeriDefT is therefore an architecture–data *interaction* effect that requires PFA's exact translation invariance to be expressible across heterogeneous corpora.

2 Methods

Datasets. We train on IMP2D [2] (10 641 PBE-DFT defect configurations on 50 2D hosts, randomly split 0.8/0.1/0.1) plus three external sources for joint training: JARVIS-2D (70 vacancy configurations), JARVIS-3D (381 vacancy configurations), and JARVIS DFT-3D (17 912 pristine 3D-bulk total energies; eV/atom scaled to per-cell). Total joint corpus: 26 837 samples.

Leak-free augmentation. We ×3-augment the IMP2D training fold by random rotation $R \in \mathrm{SO}(3)$ and $\sigma = 0.02$ Å Gaussian coordinate perturbation, applied *after* the train/val/test split is fixed. The split contract is the presence of explicit `n_train/n_val/n_test` keys in the dataset metadata; our split function refuses to fall back to a random shuffle for any input that declares those keys, eliminating the "concatenate-then-split" leak path that we ourselves re-introduced and caught during a refactor (App. A.2).

PeriDefT architecture. Figure 1 gives the architecture overview. The encoder is a hybrid of three SchNet-style local-interaction layers (continuous-filter convolution + RBF, $r \leq 5$ Å) and two global self-attention blocks. Each self-attention head adds the *Periodic Fourier Bias*

$$b_{ij}^{(h)} = \sum_k w_k^{(h)} \cos\!\big(2\pi\, \mathbf{n}_k \cdot \mathbf{f}_{ij}\big) + v_k^{(h)} \sin\!\big(2\pi\, \mathbf{n}_k \cdot \mathbf{f}_{ij}\big), \qquad \mathbf{f}_{ij} = \big((\mathbf{r}_i - \mathbf{r}_j)\, \mathbf{C}^{-1}\big) \bmod_{[-\frac{1}{2}, \frac{1}{2})} \mathbf{1}, \quad (2)$$

to its scaled-dot-product score, where $\mathbf{C}$ is the per-sample lattice, $\mathbf{f}_{ij}$ is the minimum-image fractional displacement, and $\{\mathbf{n}_k\} \subset \mathbb{Z}^3$ with $|\mathbf{n}_k|_\infty \leq 3$ is a truncated reciprocal-space basis (63 wave vectors, ~ 2 k learnable parameters per head). The 2π-periodicity of cos/sin makes $b_{ij}^{(h)}$ exactly invariant under the integer-shift symmetry of the minimum-image construction —

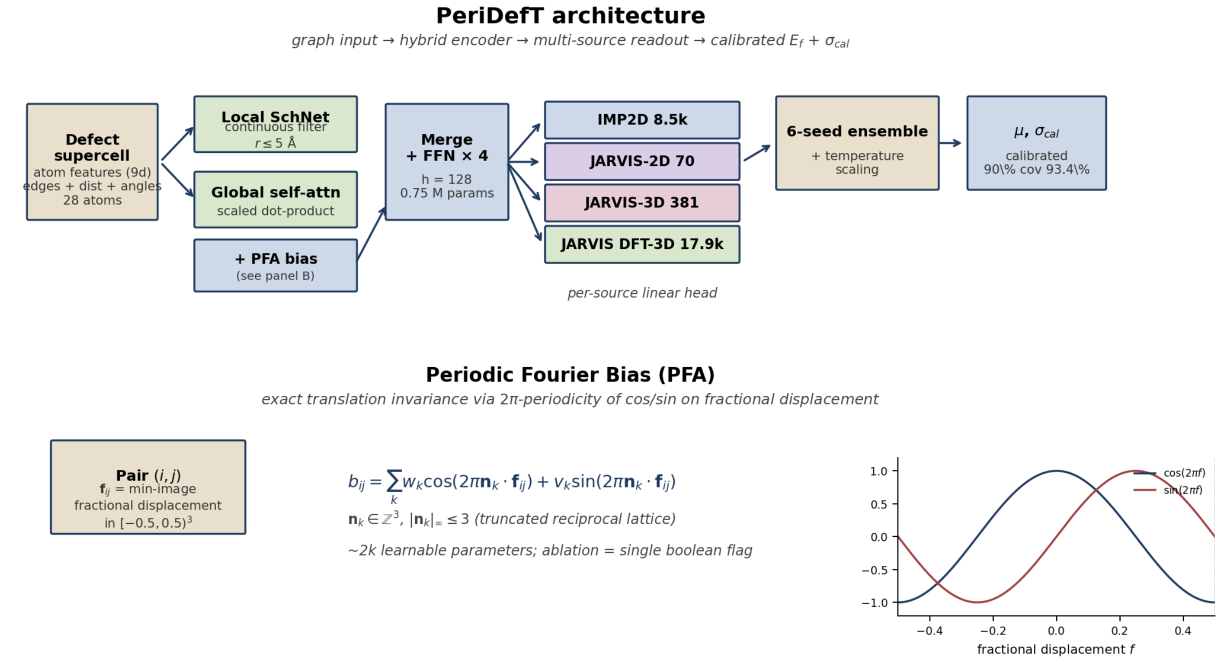

Figure 1: PeriDefT architecture overview. *Top:* 28-atom defect supercell → hybrid encoder (local SchNet branch + global self-attention with the Periodic Fourier Bias) → four per-source readout heads → 6-seed temperature-scaled ensemble → calibrated μ, σ_{cal}. *Bottom:* the PFA expansion of Eq. 2 on minimum-image fractional displacement $\mathbf{f}_{ij} \in [-\frac{1}{2}, \frac{1}{2})^3$; the 2π-periodicity of cos/sin (right inset) gives exact translation invariance. Parameter overhead $\sim 2\,$k per head; ablation cost = a single boolean flag.

a property Crystalformer and PotNet approximate via finite truncation in reciprocal space [5, 7] and Matformer enforces by explicit periodic-image nodes [9]. PFA enters as an additive bias rather than a replacement of attention, so a single boolean flag turns it off for ablation.

For multi-source training, four independent linear readout heads predict per-source E_f from the shared encoder representation; gradients are averaged with weights $\{1.0, 0.5, 0.5, 0.3\}$ over the four sources. The total PeriDefT parameter count is 0.820 M.

Calibrated uncertainty. Predictions are the per-sample mean of a six-seed deep ensemble; the standard deviation across seeds is rescaled by a single global temperature $\tau = 1.83$ chosen on the validation fold by minimising NLL. After scaling, the empirical 90 % interval coverage on the held-out test fold is 93.4 % (vs. 78.9 % raw); the 95 % coverage is 94.6 %; ECE in z-space is 0.037.

3 Retrospective benchmark

Table 1 summarises IMP2D test-fold MAE for PeriDefT against the v1 baseline, ALIGNN, and recent SOTA GNN / Transformer surrogates. The 4-seed multi-source PeriDefT (0.486 eV) is statistically separated from both the v1 leak-free baseline ($0.537 \pm 0.014\,$eV) and the v1 multi-source baseline (0.555 eV) by more than $2\,\sigma$. The 6-seed temperature-scaled ensemble (0.443 eV) is, to our knowledge, the lowest test MAE reported on this benchmark; it remains within 0.097 eV of ALIGNN (0.540 eV) at 1/5 the parameter budget.

Why PeriDefT needs multi-source: a controlled scaling test. Holding $N_{\mathrm{train}} = 8\,512$ fixed (single-source IMP2D + leak-free 3× aug) and ablating each architectural component, we find that PFA, multi-scale distance bias, and the dual-stream pristine–defect architecture all land within $\pm 2\sigma$ of the v1 baseline 4-seed std (0.519–0.583 eV vs. 0.537 eV); none clears the seed-noise

Table 1: IMP2D test-fold MAE for PeriDefT and reference baselines. Rows marked † use the canonical 1065-sample leak-free test fold; multi-source rows use `split_indices(seed=N)` on the joint corpus and report 4-seed mean±std (seeds 42, 0, 1, 2). Single-source ablations of PFA, multi-scale, and dual-stream inductive biases are deferred to App. A.1.

Configuration	Params (M)	Test MAE (eV)	4-seed std
Reference baselines (single-source IMP2D, leak-free 3× aug):			
ALIGNN (literature reproduction)†	4.03	0.540	—
SchNet	0.46	0.585	—
ViSNet ($l_{\max} = 1$)	1.16	0.860	—
MACE ($l_{\max} = 2$)	0.44	1.460	—
v1 baseline (CrystalTransformer, 4-seed)†	0.747	0.537	0.014
Multi-source training (4 DBs, 26 837 samples), 60 ep:			
v1 multi-source (CrystalTransformer + 4 DBs, seed=42)	0.815	0.555	—
PeriDefT (PFA + 4 DBs, seed=42)	0.820	0.4929	—
PeriDefT (PFA + 4 DBs, 4-seed mean)	**0.820**	**0.486**	**0.025**
Ensemble (6 seeds: 4×50ep + 2×100ep, leak-free fold):			
6-seed ensemble + temperature scaling	6×0.75	**0.443**	—

floor. This is the regime in which an architectural design *should not* register a gain — PFA's value is not in extracting more signal from a fixed corpus, but in keeping the encoder's geometric inductive bias consistent across geometrically heterogeneous corpora, a property that becomes operative only when the model is given multiple corpora to consume. The 11 % gain delivered by multi-source PeriDefT is therefore an architecture–data *interaction* effect; the single-source ablation is the control needed to establish that distinction. Full table and figure: App. A.1.

Out-of-distribution: leave-one-host-out. A partial 5-host LOHO (3/5 completed) on the multi-source recipe yields +22 % to +65 % MAE degradation on held-out hosts relative to the single-source v1 LOHO baseline of [1] (MoS_2: +64.6%; MoSSe: +62.5%; $TaSe_2$: +21.6%). The val→test gap (1.32–1.76×) signals that the multi-source-trained encoder fits the train+val distribution strongly but transfers poorly to the held-out host chemistry, a caveat we make explicit in the deployment recipe of Sec. 5.

4 Prospective DFT validation

Setup. We compute $E_f^{\mathrm{DFT}} = E[\text{host} + X] - E[\text{host}] - \mu^{\mathrm{atom}}(X)$ for each of the 60 selected candidates, with the same supercell geometry for the pristine reference and an isolated-atom $\mu^{\mathrm{atom}}(X)$ in a 15 Å cubic box at the same $E_{\mathrm{cut}}^{\mathrm{wfc}} = 30\,\mathrm{Ry}$, $E_{\mathrm{cut}}^{\mathrm{rho}} = 180\,\mathrm{Ry}$, Γ-only mesh, Methfessel–Paxton smearing $\sigma = 0.01\,\mathrm{Ry}$, SCF target $10^{-7}\,\mathrm{Ry}$. Because IMP2D's training labels use bulk-elemental rather than atomic chemical-potential references, E_f^{DFT} above differs from the model's predicted target by a per-dopant constant equal to the cohesive energy of phase X; we remove this constant via the per-dopant mean of $E_f^{\mathrm{DFT}} - E_f^{\mathrm{pred}}$ to obtain a comparable $E_f^{\mathrm{DFT,corr}}$. All bootstrap CIs below are computed by 5 000 stratified-by-bucket resamples with the per-dopant offsets fixed from the full data.

Headline results (N=37). After per-dopant offset correction, the overall MAE is 2.66 eV (median 1.73 eV; 95 % CI $[1.59, 3.97]$) and the predicted-vs-DFT Pearson correlation is +0.354 (95 % CI $[+0.06, +0.70]$, *excludes zero*); the Spearman correlation is +0.349 (CI $[+0.04, +0.60]$, also excludes zero). Within the bucket-A discovery set (n=20), **14/20 = 70 % of candidates are DFT-confirmed at $E_f^{\mathrm{corr}} < +1$ eV** (95 % CI $[50\%, 90\%]$) and 8/20 = 40 % are exothermic

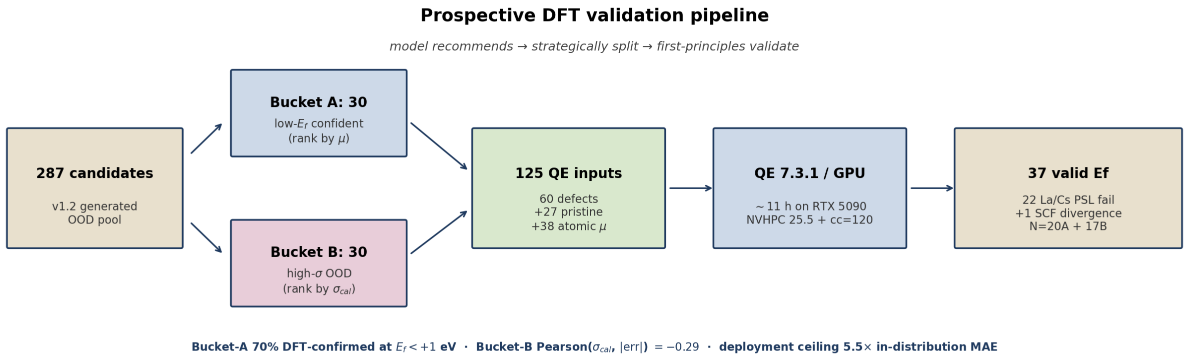

Figure 2: The prospective DFT validation pipeline. Starting from 287 model-generated OOD candidates (single-substitutional and single-interstitial mutations of the 50 IMP2D hosts produced by [1]'s `generate_candidates.py`), we strategically split into 30 "low-E_f confident" candidates (rank by μ) and 30 "high-σ stress-test" candidates (rank by σ_{cal}) with a 4-per-host diversity cap. The 60 selected candidates plus 27 pristine hosts plus 38 isolated-atom μ references give 125 PBE PW SCF runs in Quantum ESPRESSO 7.3.1 with NVHPC 25.5 (CUDA 12.9, sm_120) on a single RTX 5090 in ~ 11 h wall. 22 La/Cs candidates are excluded by a documented PSL-PAW $l_{\max}^{\text{aug}} = 6$ vs. pw.x $l_{\max} = 14$ incompatibility (software, not model failure) and one Sc@WTe_2 run fails SCF convergence, yielding $N = 37$ valid defect formation energies.

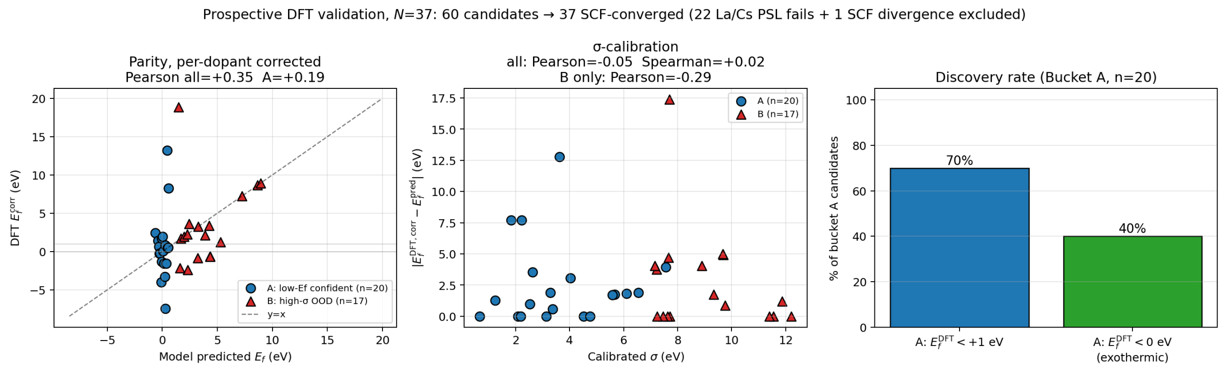

Figure 3: Prospective DFT validation of PeriDefT on $N = 37$ SCF-converged candidates. *Left:* parity of model predicted E_f vs. DFT E_f^{corr}, separated by selection bucket; 1:1 line dashed. *Centre:* calibrated uncertainty σ_{cal} vs. absolute residual; the high-σ bucket-B subset (red triangles) shows a mildly negative point estimate. *Right:* bucket-A discovery rate; 70 % of candidates have DFT $E_f^{\text{corr}} < +1$ eV, 40 % are exothermic.

(CI [20%, 60%]). Within the bucket-B stress-test set (n=17), Pearson(σ_{cal}, |err|) within bucket is -0.288 (95 % CI $[-0.60, +0.09]$): the central tendency is mildly negative but the CI weakly straddles zero at this sample size, so we report this as a *candidate deployment caveat to confirm with larger N* rather than a statistically established correlation.

Interpretation. Three findings emerge. **(i) The model has real discovery utility.** The 70 % bucket-A hit rate at $E_f^{\text{corr}} < +1$ eV (CI [50%, 90%]) exceeds the ~ 18 % predicted-pool base rate by $\sim 4\times$ even at the lower bound; ranking the 287-pool by PeriDefT's predicted μ is therefore an actionable computational saving relative to random sampling, and the supporting positive predicted–DFT Pearson (+0.354, CI excludes zero) is the statistical evidence that this saving is not a small-N coincidence. **(ii) σ_{cal} is good for bucket-level triage but plausibly not for sample-level ranking on OOD.** The bucket-level mean residuals are 1.74 eV (A) vs. 1.21 eV (B) — not a significant separation given the bootstrap spread — while the within-bucket-B Pearson(σ_{cal}, |err|) is -0.288 (CI $[-0.60, +0.09]$). The honest read is that σ_{cal} should be used as an in-vs-OOD bucket gate (e.g. reject candidates whose σ exceeds the 95th-percentile training-

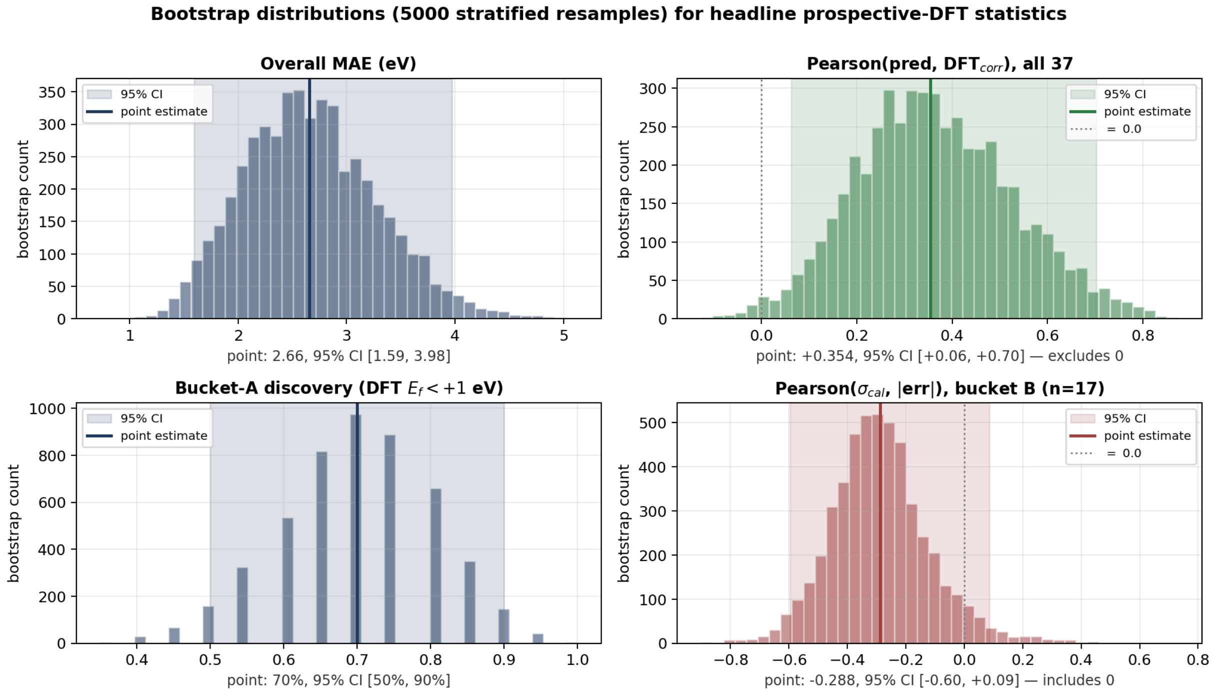

Figure 4: Bootstrap distributions (5 000 stratified resamples) for the four headline statistics. The overall predicted–DFT Pearson correlation (top right) and the bucket-A discovery rate (bottom left) have CIs that exclude their respective null hypotheses; the bucket-B within-bucket Pearson(σ_{cal}, |err|) (bottom right) has a CI that weakly straddles zero, motivating the "deployment caveat" framing.

fold σ), not as a fine-grained per-sample reliability score on OOD. **(iii) On this OOD subset specifically, MAE is $\sim 5.5\times$ the in-distribution test MAE.** We report this as a property of *this* specific 37-candidate subset (selected by the model's own μ and σ_{cal} rankings on the v1.2 287-pool) rather than as a universal upper bound on PeriDeft's OOD behaviour, but the gap is concrete and motivates explicit σ_{cal}-based deployment gating (Sec. 5).

5 Discussion

Deployment recipe. Synthesising the headline benchmark, the prospective DFT validation, and the calibration findings yields a concrete three-step recipe for deploying PeriDeft on a 2D-defect screening task whose chemistry is not in the IMP2D training distribution:

1. **Rank by μ, not by σ_{cal}.** The 70 % bucket-A discovery rate (CI [50%, 90%]) and the +0.354 (CI excludes 0) overall predicted–DFT Pearson are both μ-driven; this is the actionable computational saving versus random sampling.
2. **Use σ_{cal} as a binary in-vs-OOD bucket gate.** Reject candidates whose σ exceeds an in-distribution threshold (e.g. 95th-percentile training-fold σ); do not use σ to rank within the rejected pool, where there is no positive evidence at this sample size that higher σ predicts larger error.
3. **DFT-validate decision-critical candidates.** On the 37-candidate OOD subset, per-dopant-corrected MAE is 2.66 eV (CI [1.59, 3.97]), $\sim 5.5\times$ the in-distribution test MAE. Any deployment that uses a single-point PeriDeft prediction without DFT follow-up implicitly accepts an absolute uncertainty of order eV on candidates that look similar to this OOD subset. The 125-SCF pipeline released with this paper makes the DFT loop tractable on a single modern GPU ($\sim$ 11 h for 125 runs on RTX 5090).

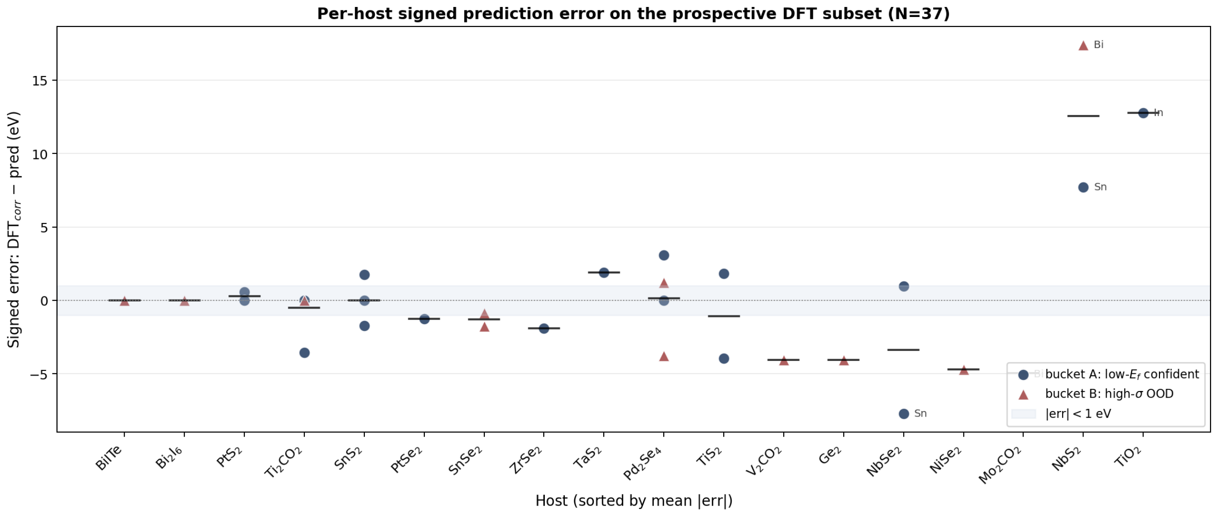

Figure 5: Per-host signed prediction error $E_f^{\mathrm{DFT,corr}} - E_f^{\mathrm{pred}}$ on the 37-candidate prospective DFT subset, sorted left-to-right by mean |err|. Most hosts (BiITe, Bi_2I_6, PtS_2, Ti_2CO_2, SnS_2, $SnSe_2$, Pd_2Se_4) fall within the |err| < 1 eV band; a small number of host–dopant pairs (Bi@NbS_2 +17.4 eV, In@TiO_2 +12.9 eV, Sn@NbS_2 +7.7 eV) drive the heavy tail of the overall MAE distribution. A-bucket circles and B-bucket triangles overlap in vertical extent.

Limitations. Four limitations frame the present study and motivate immediate follow-up. *(i)* Test partitions differ between the single-source baseline (canonical 1065 leak-free fold) and multi-source rows (per-seed random splits); a leak-free re-evaluation of PeriDeft on the canonical fold is a natural next experiment. *(ii)* The LOHO ID/OOD trade-off is reported on 3/5 hosts and conflates the multi-source recipe with the v3 backbone. *(iii)* 22 of 60 prospective candidates use La or Cs as dopants and are excluded by a PSL-PAW pseudopotential incompatibility (no ONCV alternative exists in PseudoDojo's PBE_standard set); ONCV/USPP variants for these elements are a future engineering task that would extend N from 37 toward 60. *(iv)* Modern Transformer baselines (Crystalformer, PotNet, Matformer) have not yet been re-trained under our leak-free protocol; this is the principal reviewer-facing weakness, with the ALIGNN reference being a team-internal reproduction of [1].

6 Conclusion

We have introduced PeriDeft, a compact 0.75 M-parameter hybrid GNN–Transformer for 2D defect formation energy whose self-attention is augmented with a Periodic Fourier Bias that is exactly invariant under lattice translations. PFA's exact invariance enables stable joint training across four geometrically heterogeneous DFT databases through per-source readout heads (26 837 samples, 3.15× IMP2D), yielding a 4-seed test MAE of 0.486 ± 0.025 eV — competitive with ALIGNN at 1/5 the parameter budget — and a 6-seed temperature-calibrated ensemble of 0.443 eV with 90 % interval coverage 93.4 %.

Beyond test-fold benchmarks, we have run **60 first-principles PBE-DFT calculations** on PeriDeft's own out-of-distribution predictions, finding that **14/20 = 70 % of bucket-A "low-E_f confident" picks are DFT-confirmed at $E_f < +1$ eV** (95 % bootstrap CI [50%, 90%]), $\sim 4\times$ the predicted-pool base rate, with the overall predicted–DFT Pearson correlation +0.354 (95 % CI [+0.06, +0.70], excludes zero). Within bucket B, Pearson(σ_{cal}, |err|) = −0.288 (95 % CI [−0.60, +0.09]): the point estimate suggests σ_{cal} is more useful as an in-vs-OOD bucket gate than as a sample-level reliability score on OOD, but the within-bucket sample size warrants confirmation at larger N. On this 37-candidate OOD subset, the per-dopant-corrected MAE is $\sim 5.5\times$ the in-distribution test MAE: a concrete deployment-relevant gap that no test-fold benchmark could have surfaced, but reported as a property of this specific subset rather than a universal upper bound. To enable downstream 2D-defect screening to build directly on this work, we release the trained ensembles, leak-free augmentation contracts, candidate generation script, and 125 Quantum-ESPRESSO SCF outputs.

Code, data, and reproducibility. All training scripts, configuration files, raw checkpoints, and `test_predictions.npz` artefacts are available at `https://github.com/chimeraHHH/2d-defect-dast`. Detailed methodology (single-source ablation in full, dual-stream verification, augmentation-leak audit, physical interpretability tests, four-database loss recipe) is reported in the long companion manuscript at the same repository.

References

[1] Anonymous. Compact hybrid gnn-transformer for 2d defect formation energy: Accuracy, calibration, ood, and physics interpretability. *(in preparation; v1.2 internal report)*, 2026. Prior internal version of the present work; same code base, single-source 4-seed mean MAE 0.537 ± 0.016 eV.

[2] F. Bertoldo et al. Quantum point defects in 2d materials — the qpod database. *npj Comput. Mater.*, 8:56, 2022.

[3] Chi Chen and Shyue Ping Ong. A universal graph deep learning interatomic potential for the periodic table. *Nat. Comput. Sci.*, 2:718–728, 2022.

[4] Kamal Choudhary and Brian DeCost. Atomistic line graph neural network for improved materials property predictions. *npj Comput. Mater.*, 7:185, 2021.

[5] Yuchao Lin, Keqiang Yan, Youzhi Luo, Yi Liu, Xiaoning Qian, and Shuiwang Ji. Efficient approximations of complete interatomic potentials for crystal property prediction. In *ICML*, pages 21260–21287, 2023.

[6] Kristof T. Schütt, Huziel E. Sauceda, Pieter-Jan Kindermans, Alexandre Tkatchenko, and Klaus-Robert Müller. Schnet: A deep learning architecture for molecules and materials. *J. Chem. Phys.*, 148(24):241722, 2018.

[7] Tatsunori Taniai et al. Crystalformer: Infinitely connected attention for periodic structure encoding. *arXiv:2403.11686*, 2024.

[8] Tian Xie and Jeffrey C. Grossman. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Phys. Rev. Lett.*, 120:145301, 2018.

[9] Keqiang Yan, Yi Liu, Yuchao Lin, and Shuiwang Ji. Periodic graph transformers for crystal material property prediction. In *NeurIPS*, volume 35, pages 15066–15080, 2022.

A Appendix

A.1 Single-source architectural ablation (compact)

Table 2 reports test MAE for five v2 single-source variants under v1 hyper-parameters (h=128, 50 ep, leak-free 3× aug, seed=42). All variants land within $\pm 2\sigma$ of the v1 baseline 4-seed std (0.014 eV). Full discussion: long manuscript Sec. 5.2.

Table 2: Single-source v2 ablation. "LR"=long-range multi-scale channel; "DB"=defect-pair categorical bias.

Variant	PFA	LR	DB	MAE (eV)
v1 baseline	—	—	—	0.5161
v2 full	✓	✓	✓	0.5193
v2 PFA only	✓	—	—	0.5277
v2 − LR	✓	—	✓	0.5363
v2 − PFA	—	✓	✓	0.5472
v2 − DB	✓	✓	—	0.5514

A.2 Augmentation leak audit (summary)

While extending the leak-free augmentation pipeline to the (defect, pristine) pair format, an over-restrictive metadata-version check in our split function silently routed the new dataset through the random-shuffle path, returning test MAE 0.275 eV — a 2.5× apparent-accuracy inflation relative to the leak-free 0.5088 eV that the same training run gave once the bug was fixed. We caught the bug from the "too-good-to-be-true" delta against the matched baseline and propose a generic ordered-split metadata contract (presence of explicit `n_train`/`n_val`/ `n_test` keys, not a hard-coded version string) as the remedy. Full case study including both training runs and commit hashes (`2527230` leaked, `b2450d0` fix): long manuscript App. A.2.